LV2 INSTRUMENT FOR THE ISLA S2400 DSP CARD

KeySampler

Record a sound from the S2400 into a DSP bus and play it chromatically from a MIDI keyboard.

Plugin version	0.40 (LV2 minor version 40)
Built for	S2400 DSP Card firmware 3.1.12
Controller preset	Edirol PCR-M30/M50/M80, GM2 set (memory 0)
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KeySampler is an independent plugin and is not made or supported by ISLA Instruments or Roland. Page references marked "S2400 manual" refer to the ISLA S2400 User Manual (July 2026); "DSP manual" refers to the S2400 DSP Card User Manual (March 2025); "PCR manual" refers to the PCR-M30/M50/M80 Owner's Manual.

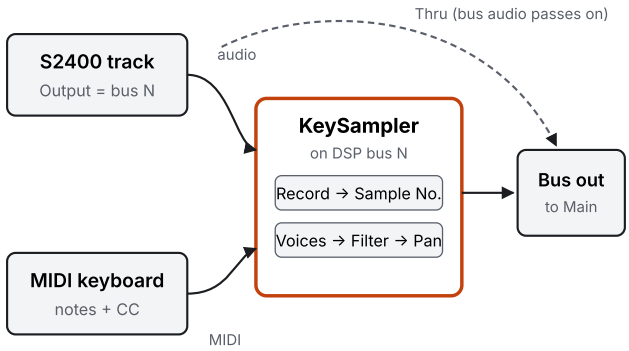
1 Overview

KeySampler sits on a DSP bus like any other plugin. It listens to the audio that the S2400 sends into that bus, records it when you arm it, and then plays the recording across the keyboard: the **Root Note** plays the sound at its original pitch, keys above and below transpose it.

Features

- Records the bus input; starts automatically when sound arrives, stops after 0.4 s of silence (max 30 s)
- Every recording gets its own number (1–9999), saved with the S2400 project
- 16-voice polyphony, velocity, pitch bend, sustain pedal, portamento and legato glide
- One Shot, Gate, Loop and Ping-Pong play modes with start and loop points in the S2400's own time format
- ADSR envelope
- Built-in sine, triangle, square and sawtooth oscillators (Sample 0–3)
- Moog-style 24 dB/oct ladder filter per voice (low, band, high pass) with drive, self-oscillation and its own ADSR envelope
- Four LFOs with delay: volume (tremolo), filter (wah), pitch (vibrato) and pan (auto-pan); pan, expression
- MIDI CC control laid out for the Edirol PCR-M30 GM2 map
- **Thru** keeps the bus passing its normal audio

Signal flow



The track's audio goes into KeySampler's bus. Thru passes it on unchanged; the filter and pan only affect KeySampler's own notes.

Where sounds come from. The DSP Card cannot read the S2400's SD card or sample memory. It only receives parameter values and, through *Sync Files*, the contents of `SD:/dspcard/`. KeySampler therefore gets its sounds by recording the bus input, or from WAV files placed in its own `samples` folder.

2 Installation

What you need

- An S2400 with the ISLA DSP Card installed
- A computer to copy one folder to the SD card (only for installing or updating the plugin)
- The plugin folder `keysampler.lv2`, containing `keysampler.so`, `keysampler.ttl`, `manifest.ttl` and a `samples` folder

Copy the plugin to the SD card

1. Put the S2400's SD card in your computer.
2. Copy the `keysampler.lv2` folder into `dspcard/plugins/lv2/` on the card (the folder where your other DSP plugins are). It must sit directly inside `lv2`, not in an extra sub-folder:

   ```
   SD card / dspcard / plugins / lv2 / keysampler.lv2 / keysampler.so
   SD card / dspcard / plugins / lv2 / keysampler.lv2 / manifest.ttl  …
   ```

3. When updating, delete the old `keysampler.lv2` folder first so no old files remain.
4. On a Mac, remove the hidden files Finder adds, then eject the card (replace `SDNAME` with the card's name):

   ```
   find /Volumes/SDNAME/dspcard -name ".DS_Store" -delete
   dot_clean /Volumes/SDNAME
   ```

Transfer it to the DSP Card

1. Put the SD card back in the S2400.
2. Press **Shift+Effects** (8th key in the top row), scroll to the **DSP Card** section and select **Sync Files** (DSP manual p. 6). Wait until it finishes.
3. **Switch the S2400 off and on again.** The DSP Card only builds its plugin list at start-up.

Load it on a bus

1. Press **Bank+0** to open the DSP Mixer.
2. Press **Shift+Pad** on a bus to open Bus Settings, select **+Add Plugin/Preset** and choose **KeySampler** (listed under the vendor *KeySampler*).
3. Highlight KeySampler and press **Enter** to open its parameters.

> **32 parameters in the menu.** The S2400 shows only the first 32 parameters of a plugin. KeySampler puts the 32 most-used ones there (chapter 10); the rest keep sensible defaults and several of them can be changed from a MIDI controller.

> **After an update** that adds or changes parameters, remove KeySampler from the bus and add it again, then re-enter its settings in each project that uses it. Recordings are not affected. **0.36, 0.38, 0.39 and 0.40 reorder the parameters** (0.40 replaces Thru and Filter Env in the menu with MIDI Channel and F.Env Amount): a project saved with an earlier version loads its values into the wrong places. Use **Changes → Default** (it keeps the sound), set it up again and save the project.

3 Quick start

This records a pad and plays it from the keyboard in about a minute.

1. **Choose the bus.** Load KeySampler on bus 1 (see *MIDI routing* below).
2. **Route the pad to it.** Hold **Shift** and press the pad you want to sample to open Track Settings (S2400 manual p. 56). Set **Output** to **1**. Hit the pad: you hear it through KeySampler because **Thru** is on.
3. **Record.** Tick **Record** in the menu, hit the pad once, then untick it.
4. **Play.** KeySampler gives the recording a new number and selects it: **Sound** shows **Sample** when you re-open the page. Play the keyboard. Middle C (note 60, the default **Root Note**) plays the sound at its original pitch.

Routing a track to a bus

With the DSP Card installed, each of the S2400's eight outputs feeds the DSP bus with the same number (DSP manual p. 3). A track's **Output** in Track Settings therefore selects the bus. Stereo samples play through an output pair: a stereo sound set to output 1 or 2 uses buses 1 and 2. Make that pair stereo in Bus Settings (**Mix**), or set the track's **Output Mix** to *L+R Mixed* to keep it on one bus (S2400 manual p. 56). In Mixer fader mode the output number of each track is shown above its fader (S2400 manual p. 25).

MIDI routing

The DSP Card gives each bus its own MIDI channel (from the card's configuration): bus *n* receives channel *n*, a stereo pair receives the channel of its first bus (1+2 → 1, 3+4 → 3, ...), and **every bus also receives channel 16**. With **MIDI Channel** on **Omni** (the default) KeySampler plays notes on any channel it is given. So to play KeySampler on bus 3, set the keyboard (or MIDI track) to channel 3, or to channel 16 — but channel 16 reaches every KeySampler at once.

Set **MIDI Channel** (Omni, 1–16) to make KeySampler listen to one channel only; notes, controllers, pitch bend and program changes on other channels are ignored. Only channels the DSP Card sends to this bus can arrive (its own number and 16), so pick one of those: for example, set it to the bus number so the shared channel 16 no longer plays it. If the keyboard goes silent, set it back to Omni.

> **Keep the bus to yourself while recording.** Everything that plays into KeySampler's bus is recorded, so route only the pad you are sampling to that bus during a take.

4 Recording

KeySampler records whatever arrives at its bus input. Each recording is stored as a new sound with its own number; nothing is ever recorded over.

How a recording starts and ends

Stage	What happens
Armed	Turning **Record** on arms the recorder. Nothing is stored until the input is louder than **Rec Threshold dB** (default −50 dB).
Recording	Starts at the first sample above the threshold, including 32 samples (about 0.7 ms) captured just before it, so the attack is not cut.
Automatic stop	After 0.4 s below the threshold, or when 30 s have been recorded.
Manual stop	Turning **Record** off ends a recording immediately.
Clean-up	Trailing silence is trimmed to 50 ms after the last loud sample and the end gets a short fade-out.
Saved	The recording gets the next free number and is written to the DSP Card as `sample_NNNN.wav` (16-bit stereo, 48 kHz). KeySampler plays it at once, sets **Sound** to **Sample** and keeps the number in the hidden **Sample ID** parameter, so saving the project keeps it.

To record another take, turn **Record** off and on again. Turning it on while a recording is being saved has no effect.

Threshold

- If background noise or a reverb tail keeps the recording running (the new sound only plays once it ends), raise **Rec Threshold dB**, or turn Record off to end the take.
- For very quiet sounds, lower it towards −80 dB.

Thru

Thru (on by default; a hidden parameter, not in the S2400 menu) passes the bus input to the bus output unchanged, so the bus behaves normally. With Thru off, the bus only outputs KeySampler's notes. Thru does not affect recording.

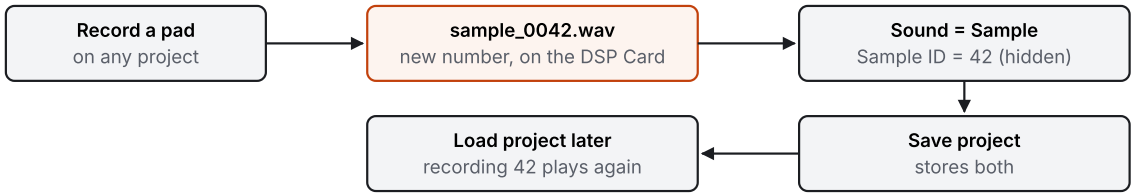
> **Mono buses** record mono; a stereo bus pair records stereo. The recorded level is the bus input level before KeySampler's Level, filter and pan.

5 Sample numbers and projects

Sound chooses what KeySampler plays. Its menu lists, in order:

Sound	Plays
Sine, Triangle, Square, Saw	The built-in oscillators. They are part of the plugin, so they can never be recorded over or deleted. New KeySamplers start on Sine, a handy test that notes arrive.
Sample	This KeySampler's sound: your latest recording, or the last recording you picked by number, whichever came last. After you record, Sound shows **Sample**. The number is kept in a hidden parameter, **Sample ID**, that the S2400 saves with the project like any other. Before the first recording or pick, Sample is silent.
1 – 9999	Any recording on the DSP Card by its number, including ones made in other projects. Picking a number that exists also makes it this KeySampler's **Sample**.

Every recording gets the next **unique number** and is stored on the DSP Card under that number, as `sample_0001.wav`, `sample_0002.wav` … Because Sound and Sample ID are saved with the project, loading it brings back exactly the sound it was saved with.



A recording keeps its number for good, so each project keeps its own sounds.

Situation	Behaviour
New recording	Gets one more than the highest number ever given out, even if that recording was deleted, so an old project never finds someone else's sound under its number. After 9999, deleted numbers are reused, lowest first. Existing sounds are never overwritten.
Choosing a sound	Set **Sound** to a number to play that recording, and leave it there to use it in this project (the project then saves the number). One sound at a time per KeySampler; switching stops notes still sounding.
Unused number	Silence.
Storage	On the DSP Card's internal storage. Recordings survive power-off, Sync Files and plugin updates. They are not on the SD card, so they do not move with a project to another S2400.

> **After recording,** re-open KeySampler's page: Sound should read **Sample**. Then save the project. If it still shows the old value, the S2400 did not receive it (this happens only with two KeySamplers whose settings are all identical): change any setting on one of them and record again.

Oscillators

The oscillators make KeySampler a simple synth. Root Note plays middle C (261.63 Hz); the waveforms are band-limited per octave, so high notes stay clean. They sustain while a key is held and stop on release whatever the Mode; Start and loop points don't apply. Everything else (envelopes, filter, vibrato, glide, CC control) works as usual. Recording works on any Sound setting and switches to the new recording (Sound = Sample).

Deleting recordings

The DSP Card has no file browser, so KeySampler deletes its own recordings:

1. Scroll **Sound** past Sine … Saw and Sample to the **number** of the recording you want to remove, and play it to be sure.

2. In **Changes**, select **Delete**.

The recording is removed from the card at once, Sound goes back to **Sample**, and Changes goes back to **---**. Because picking the number also made it the Sample, Sample returns to the one it played before you picked it. Projects that used the deleted number play silence. There is no confirmation, but Delete only works on a sound picked by number: with Sound on **Sample** or on a waveform it does nothing, so the recording a project is using can't be deleted by accident. WAVs added with Sync Files are left alone (remove those from the SD card and sync).

Recordings from earlier versions

Versions up to 0.18 stored recordings by Bank and Slot. The first time 0.20 starts, it renames them: **Bank *b*, Slot *s* becomes Sample *b*×100 + *s*** (Bank 1 Slot 3 → 103, Bank 12 Slot 5 → 1205). Recordings made with 0.19 keep their numbers. Open each older project, set Sound to the new number and save it again.

From 0.20–0.31 to 0.32: those versions named files after the internal Sample value, so the first recording was `sample_0004.wav` and showed as “4”. The first time 0.32 starts it renames every recording to the number the menu now shows (`sample_0004.wav` → `sample_0001.wav` , `sample_0027.wav` → `sample_0024.wav`) and leaves a file `numbered_from_1` so it only does this once. Projects keep playing the same sound; only the label changes (“27” now reads “24”). The Sample range grew by three, so in a project saved with a recording above about 1660 the value may come back one or two numbers off: check it and save again.

From 0.32 and earlier to 0.33: the Sound list gained the **Sample** entry, so a project saved with a recording number now shows the next lower one (a project on “24” plays recording 23). Open it, set Sound to the right number (or record again) and save. File names do not change.

Adding WAV files with a computer

Name a WAV `sample_NNNN.wav` (four digits, e.g. `sample_9000.wav`), copy it into `SD:/dspcard/plugins/lv2/keysampler.lv2/samples/` and run **Sync Files**; Sound 9000 then plays it. A recording with the same number takes priority, so use numbers your recordings won't reach (e.g. 9000 and up). Supported: 8/16/24/32-bit PCM and 32/64-bit float, mono or stereo, any sample rate; loop points stored in the WAV (smpl chunk) are used in Loop mode.

6 Playing, envelope and sample points

Playing

Parameter	Description
Level	Output volume of KeySampler's notes, 0–2 (1 = unity).
Root Note	The note that plays the sound at its original pitch (default 60 = middle C).
Tune	±24 semitones, fractions allowed (0.5 = 50 cents).
Bend Range	Pitch-bend range in semitones (0–24, default 2). Not in the S2400 menu.
Vel Sens	Not in the S2400 menu (default 1 = full velocity response). 0 = all notes at full volume. Gain = (1 − s) + s × (velocity/127)².
Voices	Polyphony limit, 1–16. Set 1 for mono lines. When all voices are busy, releasing voices are reused first, then the oldest. Replaying a held note retriggers it.
Mode	**One Shot** plays to the end and ignores note-off (drums). **Gate** fades out with Release when the key is let go. **Loop** repeats the loop section while the key is held, then fades out. **Ping-Pong** does the same but plays the loop section forwards and backwards in turn, so the loop never jumps.

The sustain pedal (CC64) holds notes in Gate, Loop and Ping-Pong modes.

Glide: portamento and legato

Glide Mode	Behaviour
Off	Every note starts at its own pitch.
Portamento	Every new note slides from the previous note's pitch to its own, also when the previous key was already released. Works with any number of Voices; with Voices = 1 the slide starts from wherever the sounding note currently is.
Legato	Mono. While a key is held, a new key does not restart the sound or the envelope: the pitch slides to the new note. Releasing it slides back to the most recent key still held. A note played with no key down starts normally, without a slide.

Glide ms (0–2000, default 100) is the time a slide takes, whatever the interval. In Legato, 0 gives instant pitch changes without retriggering, the classic "hammer-on" feel.

Envelope

Attack ms (0–2000), **Decay ms** (0–5000), **Sustain** (level 0–1) and **Release ms** (0–5000) shape each note's volume. Attack and decay are linear ramps; release fades from the current level. With Sustain 0 in Gate, Loop or Ping-Pong mode, a note ends after its decay. These four are hidden parameters, set with sliders S1–S4 (with Modulation on anything but Filter) and saved with the project.

Start and loop points

Each point uses two whole-number parameters, **sec** and **ms**, matching the S2400's waveform editor, which shows points as minutes:seconds.milliseconds+samples (S2400 manual p. 30). The sample-accurate "smp" parts exist but sit beyond the S2400's 32-parameter menu, so in the menu points move in 1 ms steps.

S2400 shows	KeySampler parameters
`0:01.250+24`	**sec** = 1, **ms** = 250 (the +24 samples are dropped)
`0:00.500+0`	**sec** = 0, **ms** = 500

- **Start** sets where every note begins.
- **Loop Start** and **Loop End** set the section repeated in Loop and Ping-Pong mode.
- Loop End at 0:00.000+0 means *end of the sound*; so does any value past the end.
- With Loop Start and Loop End both at 0, a WAV's own loop points are used if it has any, otherwise the whole sound loops.
- Samples are counted at the sound's own sample rate (48 per ms for recordings).

> **Editing tips.** Each encoder click changes a whole-number parameter by one step. For big moves, highlight the parameter and use **Fader 8**, or **F1**/**F2**/**F3** for 25/50/75 % (DSP manual p. 11). A knob assigned to Start, Loop Start or Loop End sweeps across the whole sound, which is quick for finding a loop by ear before fine-tuning it here. If a loop clicks, nudge its points by a few ms or samples.

7 Filter, LFO and pan

These affect only KeySampler's notes, never the Thru signal.

Filter

Parameter	Description
Filter	Off, Low Pass, Band Pass (mid) or High Pass. A 4-pole transistor-ladder filter in the style of the Moog, on each note: 24 dB/octave, with saturation inside the feedback loop, so loud sounds and high resonance growl and distort. Band and High Pass are mixed from the ladder stages.
Cutoff	0–100, spread musically: 0 = 20 Hz, 50 ≈ 630 Hz, 100 = 20 kHz (frequency = 20 × $1000^{\text{Cutoff}/100}$ Hz). Changes glide over about 10 ms, so sweeps are smooth.
Resonance	0–1. The cutoff frequency gets louder as you turn it up; from about **0.87** the filter **self-oscillates**: it whistles a sine at the cutoff frequency on top of the sound, which you can play by sweeping Cutoff. As on a Moog, high resonance thins the bass; KeySampler gives part of it back.

Filter envelope

The filter runs separately on every note, so each note can have its own sweep. A second envelope moves the cutoff: cutoff = **Cutoff** + envelope × **F.Env Amount**. **F.Env Amount 0 switches it off** (the default). **F.Env Amount** is in the menu; the envelope's Attack, Decay, Sustain and Release are hidden parameters, set with S1–S4 while **Modulation** is on Filter.

Parameter	Description
F.Env Amount	−100 to +100, in Cutoff units; 0 = off. +60 with Cutoff 20 opens the filter to 80 at the envelope's peak; negative values close it instead.
F.Env Attack / Decay ms	Time to reach the peak (0–2000 ms) and to fall to the sustain level (0–5000 ms).
F.Env Sustain	Level held while the key is down, 0–1.
F.Env Release ms	Time to fall to zero after release, 0–5000 ms.

The filter envelope restarts with each new note, except legato notes, which keep the running sweep. It only acts when **Filter** is not Off. Typical pluck: Low Pass, Cutoff 20, Resonance 0.3, Amount +60, Attack 5, Decay 400, Sustain 0. Acid bass: Saw (Sample 3), Low Pass, Cutoff 15, Resonance 0.8, Amount +50, Decay 200, Sustain 0, Glide on.

LFOs: tremolo, filter wobble, vibrato, auto-pan

KeySampler has four LFOs, one for each section: **volume**, **filter**, **pitch** and **pan**. Each is a sine that runs separately on every note, and they can all run at once with their own speed and delay. The menu item **Modulation** (Volume / Filter / Pitch / Pan) chooses which LFO you are editing: the menu items **LFO Rate Hz**, **LFO Depth** and **LFO Delay ms** show and change that LFO, and so do knobs R3–R5. Switch Modulation and re-open the page to see the other LFO's values; each LFO keeps its own settings (saved with the project as hidden parameters). With Modulation on **Filter**, sliders S1–S4 set the filter envelope too (Pitch and Pan have no envelope, so there S1–S4 stay on the volume envelope). Buttons B4, B5 and B6 change Modulation the same way (chapter 9).

Parameter (each LFO)	Description
… LFO Rate Hz	Speed, 0.1–12 Hz (default 5). R3.
… LFO Depth	0–100 (default 0 = off). R4. **Volume:** 100 dips fully to silence, 50 to half. **Filter:** up to ±30 Cutoff units (±3 octaves) around Cutoff; Filter must be on. **Pitch:** up to ±100 cents. **Pan:** 100 sweeps from full left to full right, around the Pan setting.
… LFO Delay ms	Time after note-on before that LFO starts, 0–2000 ms; it then fades in over 200 ms. R5.

Pan and expression

Pan (−1 left to +1 right) is a balance control (the pan LFO moves each note around it): the centre leaves both sides at full level. Listen on the main mix (Mix Out, headphones, USB main): every DSP bus, mono or stereo, reaches the main mix in stereo, but an individual output jack carries only the left channel of a mono bus. On a

stereo pair (set **Mix** in Bus Settings, DSP manual p. 9) output jacks 1&2 carry left and right. The expression pedal (CC11) scales KeySampler's output from 0 to full; it starts at full.

8 MIDI control, Save and Restore

Many parameters can be controlled from MIDI controllers. The assignments follow the PCR-M30 GM2 set (chapter 9, appendix A) and are the same in every project.

Save and Restore

Moving a knob, slider or button changes **only the sound**. The S2400 menu keeps its values, and those menu values are what the project saves. You decide what happens to your changes, from the keyboard or from the menu:

Command	What it does
Save menu **Changes** → Save	Writes every value changed from a controller into the S2400 menu. Save the project afterwards to keep them.
Restore menu **Changes** → Restore	Forgets all controller changes: KeySampler goes back to the saved settings, i.e. the values in the menu. The sound (Sound, Sample ID) is not changed.
Delete menu **Changes** → Delete	Deletes the recording picked by number in Sound (see chapter 5). Does nothing when Sound is Sample or a waveform.
Default menu **Changes** → Default	Puts every other setting back to its factory default, in the menu too. The sound itself (Sound, Sample ID), Record and Changes are not changed. Save the project to keep it; reload the project to undo it.

- Changing a value in the menu also takes over that one parameter, so the last thing touched is what you hear.
- Save sends the values to the DSP Card's audio engine (sushi), exactly as the menu does. Start and loop points are sent as sec / ms / smp; a Loop End at maximum is sent as 0:00.000+0 (end of sound).
- The S2400 shows the saved values when you re-open the plugin page. Two KeySamplers with identical settings cannot be told apart, and Save does nothing for them.
- **Changes** (the last menu item) acts about a second after you select Save, Restore, Default or Delete, then goes back to **---** (shown when you re-open the page). The short wait means passing over an entry while browsing does nothing, but there is no confirmation. A value saved in a project is never acted on when it loads.
- The hidden **Auto Menu Sync** (off) would copy every controller move to the menu instead.

How controller values are scaled

Function	Controller 0 → 127
Level	0 → 1 (full = unity gain)
Cutoff / Resonance / Sustain	0 → 100 / 0 → 1 / 0 → 1
Attack	0 → 2000 ms, curved (value^2) for finer short times
Decay / Release	0 → 5000 ms, curved
Tune / Pan	−12 → +12 semitones / left → right; centre (64) = 0 / middle
Start / Loop Start / Loop End	Beginning → end of the sound
Record / Thru / Filter Env (by MIDI)	Off below 64, on from 64
Sample / Filter Type / Mode	Sine … Saw, 1 → 9999 / 4 zones / 4 zones, in menu order
LFO Rate / Depth / Delay	0.1 → 12 Hz (curved) / 0 → 100 / 0 → 2000 ms
Glide Mode / Glide Time	3 zones (Off, Portamento, Legato) / 0 → 2000 ms, curved
F.Env Amount / A / D / S / R	−100 → +100 (centre 0) / 0 → 2000 ms / 0 → 5000 ms / 0 → 1 / 0 → 5000 ms

9 Using the Edirol PCR-M30

KeySampler's map follows the PCR-M30/M50/M80 **GM2 SET** (memory 0, the factory setting; PCR manual p. 98), with L1–L3 changed so they don't stop the S2400's sequencer (below). Knobs keep the functions printed on the GM2 template; the sliders carry the envelope and sample points; the buttons switch features.

Ctrl	GM2 template label	Message sent	KeySampler function
R1	Filter Cutoff	`CC 74`	Cutoff
R2	Filter Resonance	`CC 71`	Resonance
R3	Vibrato Rate	`CC 76`	LFO rate (volume LFO, or the section B4/B5/B6 selected)
R4	Vibrato Depth	`CC 77`	LFO depth (same)
R5	Vibrato Delay	`CC 78`	LFO delay (same)
R6	Coarse Tuning	`RPN 0/2`	**Glide ms**
R7	Fine Tuning	`RPN 0/1`	Tune, fine: ±100 cents around the menu Tune
R8	Pan	`CC 10`	Pan
S1	Portamento Time	`CC 5`	Attack (volume envelope, or filter envelope after B5)
S2	Aftertouch	`Channel pressure`	Decay (volume or filter envelope)
S3	Envelope Attack	`CC 73`	Sustain (volume or filter envelope)
S4	Envelope Decay	`CC 75`	Release (volume or filter envelope)
S5	Envelope Release	`CC 72`	Start point
S6	Chorus	`CC 93`	Loop Start
S7	Reverb	`CC 91`	Loop End
S8	Volume	`CC 7`	Level
B1	Program Change Dec	`Program change −`	Filter: Off → Low Pass → Band Pass → High Pass → Off
B2	Program Change Inc	`Program change +`	Mode: One Shot → Gate → Loop → Ping-Pong → One Shot
B3	Portamento On/Off	`CC 65 (0/127)`	Glide on / off (back to the last Glide Mode used, Portamento by default)
B4	Poly Mode On	`CC 127`	**Pitch modulation:** R3–R5 set the pitch LFO
B5	Mono Mode On	`CC 126`	**Filter modulation:** R3–R5 set the filter LFO, S1–S4 the filter envelope
B6	GM2 System On	`SysEx F0 7E 7F 09 03 F7, or CC 87`	**Pan modulation:** R3–R5 set the pan LFO. If the S2400 doesn't pass the SysEx on, set B6 to CC 87 (below).
L1–L3	Stop / Start / Continue → *set to CC 80–82*	`CC 80–82`	Not used (free for later)
P1	Hold	`CC 64`	Sustain pedal
P2	Expression	`CC 11`	Expression (output volume)
Lever	Pitch bend / modulation	`Pitch bend / CC 1`	Pitch bend (Bend Range). Modulation is not used.

- **B1 and B2** both send program changes; KeySampler tells them apart by direction (lower number = B1, higher = B2). When the PCR is already at 0 or 127 a repeated number still counts: 0 = B1, anything else = B2.
- **B4, B5, B6** set **Modulation** (also in the menu): which section the LFO knobs R3–R5 set: pitch, filter or pan. On Volume the knobs set the **volume** LFO. Press the button before (or while) turning the knobs. The choice stays until you press the **same button again** (back to Volume) or **another** of the three (that section). Like other button changes it lasts until Restore; **Changes → Save** writes it into the menu. B5 also moves sliders S1–S4 to the filter envelope; you hear them only when a Filter type is on and **F.Env Amount** is not 0.
- The PCR sends nothing when B4–B6 are released, so KeySampler can't tell how long you hold them: one press selects, the next press decides.
- Buttons B1–B3 start from the current setting, including one chosen in the menu. Like knobs, they change the sound only until you use **Changes → Save**.
- R7 fine-tunes around the menu **Tune**; changing Tune in the menu takes over again.

Setting L1–L3 on the PCR (once)

KeySampler doesn't use L1–L3 at the moment. In the GM2 set they send Stop / Start / Continue, transport messages that start and stop the S2400's own sequencer, so it's still worth setting them to control changes and saving the result as a user memory (PCR manual p. 63, 59, 84):

1. Select memory **0** (GM2). Press **EDIT**, press **L1**, then **ENTER**.
2. Press the **CONTROL CHANGE** key, **ENTER**. Enter the MIDI channel of KeySampler's bus (chapter 3), **ENTER**. Enter **80**, **ENTER**. Press **0** (Latch mode), **ENTER**.
3. Repeat for **L2** with **81** and **L3** with **82**.
4. Press **EDIT**, the **SAVE** key, **ENTER**, choose memory **1**, **ENTER**. Use memory 1 from now on; everything else is the same as the GM2 set.

If B4, B5 or B6 does nothing

B6 sends a universal SysEx message (GM2 System On), and B4/B5 send channel mode messages (CC 127/126). The S2400 may keep these for itself instead of passing them to the DSP Card. KeySampler also accepts plain control changes for the three buttons: **B4 = CC 85, B5 = CC 86, B6 = CC 87**. To set one, for example B6:

1. Select your memory (1). Press **EDIT**, press **B6**, then **ENTER**.
2. Press the **CONTROL CHANGE** key, **ENTER**. Enter the MIDI channel of KeySampler's bus, **ENTER**. Enter **87**, **ENTER**. Press **0** (Latch mode), **ENTER**.
3. Press **EDIT**, the **SAVE** key, **ENTER**, choose memory **1**, **ENTER**.

> **Check on your S2400.** Whether it forwards aftertouch (S2), program change (B1/B2), mode messages (B4/B5), RPN (R6/R7) and SysEx (B6) to the DSP bus is not confirmed. If a control does nothing, use the menu.

10 Parameter reference

All parameters in port order; the S2400 shows the first 32 (ports 5–36). Ports 0–4 are MIDI in and audio out/in L/R. The # in the S2400's .DSP file is the port minus 5.

Port	Name	Range	Default	PCR	Notes
In the menu					
5	Sound	Sine, Triangle, Square, Saw, Sample, 1 – 9999	Sine		Sample = own sound (Sample ID); internal 0–10003, number N = N + 4
6	Record	tick box	off		Ticked arms recording
7	Rec Threshold dB	−80 – −6	−50		Level that starts a recording
8	Mode	One Shot / Gate / Loop / Ping-Pong	Gate	B2	
9	Level	0 – 2	1	S8	
10	Root Note	0 – 127	60		Note at original pitch
11	Tune	−24 – 24	0	R7	Fractions allowed
12	Start sec	0 – 59	0	S5	
13	Start ms	0 – 999	0	S5	
14	Loop Start sec	0 – 59	0	S6	
15	Loop Start ms	0 – 999	0	S6	
16	Loop End sec	0 – 59	0	S7	0:0 = end of the sound
17	Loop End ms	0 – 999	0	S7	
18	Filter	Off / Low Pass / Band Pass / High Pass	Off	B1	Moog-style ladder
19	Glide Mode	Off / Portamento / Legato	Off	B3	Legato = one note at a time
20	Pan	−1 – 1	0	R8	Balance
21	Voices	1 – 16	16		Polyphony
22	MIDI Channel	Omni, 1 – 16	Omni		Omni = any channel
23	Vel Sens	0 – 1	1		Velocity response
24	Bend Range	0 – 24	2		Semitones
25	Start smp	0 – 95	0	S5	Fine-tune in samples
26	Loop Start smp	0 – 95	0	S6	
27	Loop End smp	0 – 95	0	S7	
28	F.Env Amount	−100 – 100	0		Cutoff units; 0 = filter envelope off
29	Modulation	Volume / Filter / Pitch / Pan	Volume	B4/B5/B6	Which LFO the three LFO items and R3–R5 edit (and S1–S4 on Filter)
30	LFO Rate Hz	0.1 – 12	5	R3	Of the LFO Modulation selects
31	LFO Depth	0 – 100	0	R4	Of the LFO Modulation selects; 0 = off
32	LFO Delay ms	0 – 2000	0	R5	Of the LFO Modulation selects
33	Changes	--- / Save / Restore / Default / Delete	---		Acts, then back to ---
Also shown, after Changes (the S2400 always shows the first 32)					
34	Cutoff	0 – 100	100	R1	
35	Resonance	0 – 1	0	R2	
36	Glide ms	0 – 2000	100	R6	
Hidden: saved with the project, set from the PCR, the LFO items or left at the default					
37	Thru	tick box	on		Bus input to output
38	Filter Env	tick box	on		Master switch for the filter envelope
39	Attack ms	0 – 2000	1	S1	Volume envelope
40	Decay ms	0 – 5000	200	S2	
41	Sustain	0 – 1	1	S3	
42	Release ms	0 – 5000	150	S4	
43	F.Env Attack ms	0 – 2000	5	Filter: S1	

Port	Name	Range	Default	PCR	Notes
44	F.Env Decay ms	0 – 5000	300	Filter: S2	
45	F.Env Sustain	0 – 1	0.2	Filter: S3	
46	F.Env Release ms	0 – 5000	300	Filter: S4	
47	Auto Menu Sync	tick box	off		Copy every move to the menu
48	Sample ID	0 – 9999	0		Recording Sound = Sample plays
49	Vol LFO Depth	0 – 100	0	Volume: R4	
50	Vol LFO Rate Hz	0.1 – 12	5	Volume: R3	
51	Vol LFO Delay ms	0 – 2000	0	Volume: R5	
52	Filter LFO Depth	0 – 100	0	Filter: R4	
53	Filter LFO Rate Hz	0.1 – 12	5	Filter: R3	
54	Filter LFO Delay ms	0 – 2000	0	Filter: R5	
55	Pitch LFO Depth	0 – 100	0	Pitch: R4	
56	Pitch LFO Rate Hz	0.1 – 12	5	Pitch: R3	
57	Pitch LFO Delay ms	0 – 2000	0	Pitch: R5	
58	Pan LFO Depth	0 – 100	0	Pan: R4	
59	Pan LFO Rate Hz	0.1 – 12	5	Pan: R3	
60	Pan LFO Delay ms	0 – 2000	0	Pan: R5	

11 Troubleshooting and limitations

Problem	What to check
KeySampler is not in the plugin list	Run Sync Files, then power-cycle the S2400. Check the folder is `dspcard/plugins/lv2/keysampler.lv2/` with the files directly inside, no extra sub-folder, no zip. Remove `.DS_Store` files. Look under the vendor **KeySampler** or **All vendors**.
The pad is not heard through the bus	The track's **Output** must match KeySampler's bus number; **Thru** must be on.
The keyboard plays nothing	Set **Sound** to Sine: a sine tone should sound. If it does not, set **MIDI Channel** to Omni; then check the notes reach this bus: set the keyboard to the MIDI channel with the bus number (or channel 16).
Record does nothing	Nothing reached the bus above the threshold: check the pad's Output and lower **Rec Threshold dB**.
Recorded but nothing plays	The recording may still be running (noise or a long tail keeps it open): turn **Record** off. If it never starts, lower **Rec Threshold dB**. Check that **Sound** is on **Sample** or on the recording's number.
A project plays the wrong sound	The project was saved before the recording reached the menu. Re-open KeySampler's page, check that **Sound** is **Sample**, and save the project again.
Waveforms play, recordings are silent	A Start or Loop point is past the end of the recording (oscillators ignore those points, recordings don't). Check Start / Loop Start / Loop End, or use **Changes → Default**, which resets them and keeps the sound.
Pan has no effect	The bus is mono. Make its pair stereo in Bus Settings (**Mix**).
Loop does nothing	**Mode** must be Loop. Loop End 0 means the end of the sound; check the points are inside the sound's length.
A knob does nothing	Check the PCR is on memory 0 (GM2 set) and on the bus's MIDI channel. If it still does nothing, the S2400 is not forwarding that type of message.
The menu does not follow a knob	The S2400 redraws the page only when it is opened: leave and re-open it. With two identical KeySamplers, change one setting on one of them so they can be told apart.
Settings changed after an update	Parameters may have been renamed or reordered. Remove and re-add the plugin and re-enter settings.

Known limitations

- The sound itself stays on the DSP Card; the project stores only its number.
- Recordings live on the DSP Card only (not on the SD card).
- The S2400 file browser does not show the `dspcard` folder, so adding WAV files to the `samples` folder needs a computer. Recording from the bus does not.
- After Save the S2400 shows the new values only when the plugin page is re-opened.
- Maximum recording length 30 s; maximum point position 59.999 s.

A MIDI implementation

KeySampler receives MIDI from the S2400 through its LV2 event input. It does not transmit MIDI. All messages are accepted on **every channel** (omni). Running status and system real-time messages are handled by the host and are not used.

A.1 Channel voice messages

Message	Status / data	Behaviour
Note off	`8n kk vv`	Releases the note (Gate, Loop). Ignored in One Shot. Release velocity ignored.
Note on	`9n kk vv`	kk 0–127, vv 1–127. vv = 0 is treated as note off. Pitch = kk − Root Note + Tune (semitones).
Polyphonic key pressure	`An kk vv`	Ignored.
Control change	`Bn cc vv`	See A.2–A.4.
Program change	`Cn pp`	Lower number = Filter step, higher = Mode step (A.5).
Channel pressure	`Dn vv`	Loop Start.
Pitch bend	`En ll mm`	14-bit, centre 8192. Range = Bend Range semitones. Applies to all sounding notes.

A.2 Control changes with fixed functions

CC	Name	Behaviour
`6 / 38`	Data entry MSB / LSB	Data for RPN 0/1 and 0/2 (A.4). Other RPNs and NRPNs are ignored.
`11`	Expression	Scales KeySampler's output 0–1 (initial value 127).
`64`	Hold (sustain pedal)	≥ 64 on. Notes released while held keep sounding until the pedal is released.
`98 / 99`	NRPN LSB / MSB	Ignored.
`100 / 101`	RPN LSB / MSB	Select the RPN for data entry.
`120`	All sound off	Stops all voices immediately.
`121, 122`	Reset controllers, local	Ignored.
`123`	All notes off	Releases all notes.
`124, 125`	Omni off / on	Ignored (always omni).
`65`	Portamento on/off	Button: Glide on/off, every message.
`80 / 81 / 82`	General purpose 5–7	Ignored (L1–L3, free for later).
`85 / 86 / 87`	Undefined	Buttons (value ≥ 64 = press): same as B4 / B5 / B6, for a PCR whose buttons are set to these CCs (chapter 9).
`126`	Mono mode on	Button B5: filter modulation (R3–R5 filter LFO, S1–S4 filter envelope); again = back to volume. Polyphony is set with Voices.
`127`	Poly mode on	Button B4: pitch modulation (R3–R5 pitch LFO); again = back to volume.

A.3 Control changes

Controllers mapped to KeySampler functions (other CCs are ignored):

CC	GM2 name	Function
5	Portamento time	Attack*
7	Volume	Level
10	Pan	Pan
71	Resonance (timbre)	Resonance
72	Release time	Start point
73	Attack time	Sustain*
74	Cutoff (brightness)	Cutoff
75	Decay time	Release*
76 / 77 / 78	Vibrato rate / depth / delay	LFO rate / depth / delay†
91	Reverb send	Loop End
93	Chorus send	Loop Start
Dn vv	Channel pressure	Decay*

* Volume envelope, or the filter envelope when Modulation = Filter (menu or B5). † The LFO chosen by Modulation (menu or B4 / B5 / B6); Volume by default.

Value scaling is listed in chapter 8. A controller value overrides the menu value until that parameter is changed in the menu.

A.4 Registered parameters

RPN (MSB/LSB)	Name	Behaviour
0 / 1	Fine tuning	14-bit via CC6 + CC38, centre 8192 = 0; range ±100 cents.
0 / 2	Coarse tuning	Used for **Glide ms** (R6): CC6 0–127 = 0–2000 ms, curved.
127 / 127	RPN null	Deselects; following data entry is ignored.

Fine tuning is added to the menu Tune value until Tune is changed in the menu.

A.5 Program change

Program change is used for two buttons: a number **lower** than the previous one steps the **Filter** type (B1, "program change dec"), a **higher** one steps the **Mode** (B2, "inc"). A repeated 0 counts as lower, any other repeated number as higher; the first program change received counts as lower if it is 0.

A.6 System exclusive

Message	Behaviour
F0 7E 7F 09 03 F7 (GM2 System On)	Button B6: pan modulation (R3–R5 pan LFO); again = back to volume.
All other SysEx	Ignored.

B MIDI implementation chart

Model: KeySampler 0.40 (LV2 plugin for the S2400 DSP Card) · Date: 28 October 2026 · **O** = yes, X = no
* S1–S4: volume or filter envelope (B5). Reception depends on what the S2400 forwards to the DSP bus.

Function	Transmitted	Recognized	Remarks
Basic channel			
Default	X	1–16 (all)	Omni
Changed	X	Omni or 1–16	MIDI Channel menu item
Mode			
Default	X	Mode 1 (omni on, poly); Mode 3 (omni off) with a MIDI Channel set	
Messages	X	X	CC 126/127 are used as buttons B5/B4, not as mode changes
Note			
Note number	X	0–127	
Velocity: note on	X	**O** (9n v=1–127)	Vel Sens
Velocity: note off	X	X	
Aftertouch			
Polyphonic (key)	X	X	
Channel	X	**O**	Decay (volume / filter envelope)
Pitch bend			
Pitch bend	X	**O**	0–24 semitones (Bend Range)
Control change			
5	X	**O**	Attack*
6, 38	X	**O**	Data entry (RPN)
7	X	**O**	Level*
10	X	**O**	Pan*
11	X	**O**	Expression
64	X	**O**	Hold
65	X	**O**	Glide on/off (button)
71, 72, 73, 74, 75	X	**O**	Resonance, Start, Sustain*, Cutoff, Release*
76, 77, 78	X	**O**	LFO rate, depth, delay (section chosen with B4–B6)
91, 93	X	**O**	Loop End, Loop Start
80, 81, 82	X	X	Not used
85, 86, 87	X	**O**	Pitch / filter / pan modulation (buttons, same as B4–B6)
0–119 (others)	X	**O**	Ignored
98, 99 (NRPN)	X	X	
100, 101 (RPN)	X	**O**	RPN 0/1 fine tune, 0/2 Glide ms
120, 123	X	**O**	All sound off, all notes off
126, 127	X	**O**	Buttons: 126 filter modulation (B5), 127 pitch modulation (B4)
Program change			
Program change	X	**O** (0–127)	Filter step (lower) / Mode step (higher)
System			
System exclusive	X	**O**	GM2 System On only (B6: pan modulation)

Function	Transmitted	Recognized	Remarks
System common	X	X	
System real time (incl. active sensing, reset)	X	X	
Reset all controllers, local on/off (121, 122)	X	X	

C Files and storage on the DSP Card

Location on the DSP Card	Contents
`/home/isla/user/plugins/lv2/keysampler.lv2/`	The plugin, copied from `SD:/dspcard/plugins/lv2/` by Sync Files
`…/keysampler.lv2/samples/`	Optional `sample_NNNN.wav` files added with Sync Files
`/home/isla/user/keysampler_recordings/sample_NNNN.wav`	Recordings, one number each (0001–9999), the number the Sound menu shows
`/home/isla/user/keysampler_recordings/last_number.txt`	Highest number given out so far

- Recordings are 16-bit stereo PCM WAV at the DSP Card's sample rate (48 kHz), about 190 KB per second.
- The recordings folder is outside the plugin folder, so Sync Files and plugin updates do not touch it.
- `cc_map.txt` lists `<cc> <target>` pairs that differ from the defaults (target 0 = unassigned; CC 128 = channel pressure). Deleting it, or choosing **Reset Map**, restores the defaults.

Technical data

Plugin format	LV2 instrument, aarch64, needs glibc 2.17 or newer and the urid:map feature
Ports	1 MIDI in (atom sequence), 2 audio in, 2 audio out, 42 controls
Polyphony	16 voices (1 in Legato), cubic (Hermite) interpolation; glide linear in semitones
Filter	4-pole Moog-style ladder per voice, LP / BP / HP, 20 Hz – 20 kHz, ADSR envelope
Oscillators	Sine, triangle, square, sawtooth; 2048-point tables, one band-limited table per octave (harmonics < 20 kHz)
Recording	Up to 30 s, threshold −80 to −6 dB, 0.7 ms pre-roll, 0.4 s silence stop
Storage	9996 numbered recordings (limited by the card's free space)
Save to Menu	gRPC (HTTP/2) to sushi on 127.0.0.1:51051: GetAllProcessors, GetProcessorParameters, GetParameterValue, SetParameterValue (normalized values)
Real-time safety	The audio thread never reads files or allocates memory; a background thread loads, saves and rescans the samples folder every 2 s